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GUIDING MOE REASONING VIA ROUTING INTERVENTION

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Chapter 1

Theoretical Framework

The objective of this thesis is to investigate how the reasoning behaviour of a mixture-of-experts (MoE) language model can be guided through interventions on its routing mechanism. This objective connects a behavioural question, concerning the organisation of a generated solution, with a computational question, concerning which experts process its tokens. Establishing that connection requires a vocabulary for describing reasoning, measurements of internal model activity, and a distinction between observing an association and changing the mechanism that produces it.

The project develops three experimental components toward this objective. The first constructs an automatic annotator of reasoning episodes. The second tests whether episode information is accessible from hidden states before the corresponding text is generated. The third measures associations between routing, representations, reasoning events, and answer correctness. Together, these components provide evidence for selecting possible intervention targets. They do not yet constitute an evaluated routing controller. This chapter defines the concepts and assumptions needed to interpret their outputs and to formulate the subsequent intervention problem.

1.1 Reasoning as a measurable process

Let q denote a problem and let $x_{1:T}$ be the model's generated response, including intermediate reasoning and the final answer. An autoregressive language model with parameters θ defines

$$P_\theta(x_{1:T} \mid q) = \prod_{t=1}^{T} P_\theta(x_t \mid q, x_{<t}). \tag{1.1}$$

Each generated token becomes part of the context for later decisions. Intermediate text can therefore affect subsequent computation, even when evaluation ultimately assigns a single correctness label to the final answer. Chain-of-thought prompting makes such intermediate text explicit and has demonstrated benefits on several reasoning benchmarks [1].

For analysis, the response is segmented into units $s_1, \ldots, s_J$, usually sentences or short coherent spans. A unit can restate the problem, introduce a strategy, execute a calculation, or check a previous result. These functions provide a richer description than response length or final accuracy alone. However, a verbal explanation is an observable output of the model, rather than a complete record of its internal computation. Experiments on chain-of-thought faithfulness show that a response can omit factors that influenced its answer [2]. Accordingly, this thesis uses the term *reasoning episode* for a function assigned to a text unit, without assuming that the label identifies a unique internal cognitive state.

Three levels of observation will recur throughout the thesis. At the token level, the model produces hidden representations and expert-routing decisions. At the unit level, a segment receives a functional episode label. At the attempt level, the complete response has a final answer and possibly binary indicators of particular reasoning events. Repeated attempts add a fourth level: the source problem. Keeping these levels explicit is necessary both for temporal alignment and for statistical analysis.

1.2 Sparse mixture-of-experts computation

1.2.1 Experts, routers, and conditional computation

A sparse MoE layer contains several parameterised expert networks and a router that selects a subset for each token. The sparsely gated formulation of Shazeer et al. made it possible to increase model capacity while activating only part of that capacity for an input [3]. In language-model backbones, an MoE module commonly replaces a dense feed-forward sublayer. The expert is therefore a neural transformation within a larger network; it is not an independent model that solves the whole problem.

Let $u_t^{(\ell)} \in \mathbb{R}^d$ be the input to the router at layer ℓ, and let E_ℓ be the number of routed experts there. For a softmax-based router, scores $z_t^{(\ell)}$ define

$$z_t^{(\ell)} = W_r^{(\ell)} u_t^{(\ell)} + b_r^{(\ell)}, \tag{1.2}$$

$$p_{t,e}^{(\ell)} = \frac{\exp(z_{t,e}^{(\ell)})}{\sum_{j=1}^{E_\ell} \exp(z_{t,j}^{(\ell)})}, \tag{1.3}$$

$$S_t^{(\ell)} = \mathrm{TopK}\big(p_t^{(\ell)}, k_\ell\big). \tag{1.4}$$

The bias can be absent, and other routing architectures use different score transformations. These equations specify the softmax case used to define the measurements below, rather than a universal implementation contract.

Writing $f_e^{(\ell)}$ for expert e, a routed output has the form

$$m_t^{(\ell)} = \sum_{e \in S_t^{(\ell)}} g_{t,e}^{(\ell)} f_e^{(\ell)}(u_t^{(\ell)}) + m_{t,\mathrm{shared}}^{(\ell)}. \tag{1.5}$$

Here $g_{t,e}^{(\ell)}$ denotes the effective mixing weight, which may be renormalised over the selected experts. The shared contribution is present only in architectures that include shared experts. This distinction matters: the full router distribution p, the selected set S, and the effective weights g are different objects. An intervention on one need not have the same effect as an intervention on another.

Switch Transformer uses one routed expert per token, whereas Mixtral uses two of eight experts [4, 5]. DeepSeekMoE combines finer expert partitioning with shared experts [6]. These architectural choices affect how routing statistics should be interpreted. They also illustrate why total parameter count and active parameter count must be distinguished: sparse activation reduces per-token expert computation, but does not eliminate the cost of storing expert weights or dispatching tokens.

The project's Qwen3.5-35B-A3B subject provides a concrete instance. Its official model card describes a hybrid backbone with 40 layers, combining Gated DeltaNet and gated attention, with 256 routed experts, eight selected experts per token, and a shared expert [7]. This makes the subject a hybrid sequence model. The general routing notation above applies independently of the particular sequence-mixing sublayer.

1.2.2 Interpreting expert identity

Expert identities are local to a layer. The correct identifier is the pair (ℓ, e): expert 7 in one layer has different parameters from expert 7 in another layer. Likewise, selecting an expert frequently during a labelled episode does not establish that the expert implements that episode. Token identity, syntax, position, and problem content may all influence selection. Mixtral's routing analysis, for example, reports syntactic regularities and locality between consecutive tokens [5].

It is useful to distinguish *statistical specialisation*, meaning that expert use differs across observed contexts, from *functional involvement*, meaning that changing the expert's contribution changes a behaviour of interest. The first can nominate targets for study. The second requires an intervention and an appropriate comparison. Neither implies a one-to-one mapping between experts and semantic reasoning functions.

Table 1.1: Operational episode categories used by the project, based on the adapted framework of Li et al. [9]. The descriptions concern a unit's function, rather than its mathematical validity.

Episode	Function of the text unit
Read	Restates information or the goal supplied by the problem.
Analyze	Interprets information, introduces structure, or derives a relationship.
Plan	Commits to a concrete operation or strategy before executing it.
Implement	Executes a procedure, calculation, substitution, or enumeration.
Explore	Tentatively proposes a hypothesis, option, or alternative path.
Verify	Checks an existing candidate, result, or consistency claim.
Monitor	Expresses content-light self-regulation, a transition, or a structural boundary.

1.3 A vocabulary for reasoning episodes

1.3.1 Adapting episode theory to model-generated text

Schoenfeld's account of mathematical problem solving emphasises the organisation and control of problem-solving activity in addition to the availability of mathematical knowledge [8]. Li et al. adapt this perspective to reasoning-model traces and release a corpus with sentence and paragraph annotations [9]. The present project uses seven sentence-level categories. Their operational meaning in the annotation experiment is summarised in Table 1.1.

These labels are nominal categories. They are not ordered from weak to strong reasoning, and a solution need not pass through them in a fixed sequence. A Verify unit can contain an invalid check; an Implement unit can contain an incorrect calculation. Similarly, repeated monitoring does not necessarily improve a solution. Episode analysis and correctness evaluation therefore measure different aspects of model behaviour.

Segmentation is part of this measurement. One sentence may both announce and execute a plan, while a short structural marker may have no mathematical content. Assigning one label requires a documented precedence rule and access to enough context to distinguish, for example, a given fact from a newly derived relation. These ambiguities limit the meaning of exact agreement with a reference annotation. Per-class errors and inspection of ambiguous boundaries are consequently useful alongside aggregate classification scores.

1.3.2 Automatic episode annotation

Let $a_j \in \mathcal{A}$ denote the reference episode for unit s_j, where $\mathcal{A}$ contains the seven categories. A prompted annotator with instruction ϕ returns

$$\widehat{a}_j = A_\phi(q, s_{j-1}, s_j, s_{j+1}). \tag{1.6}$$

This is the information pattern used by `gepaLLMAsJudge`: the problem and neighbouring units help identify the current unit's function. Because the current and next units are visible, it is an offline annotation tool. It cannot be inserted unchanged into a controller that must act before s_j is generated.

GEPA searches over instructions using feedback and natural-language reflection, while keeping the underlying model weights fixed [10]. In this project, a same-model judge can provide structured optimisation feedback, but final prompt selection is assessed against reference labels. Such feedback is not independent ground truth. Splitting complete responses before their units are assigned to training, validation, and test sets prevents neighbouring passages from crossing evaluation boundaries.

Accuracy measures exact label agreement. Balanced accuracy averages recall across categories, and macro-F1 averages the category-specific F1 scores; both expose failures concealed by a dominant class. Cohen's κ measures chance-adjusted nominal agreement. Rank-based agreement measures require an independently justified ordering and are not primary measures for these episode labels. The purpose of annotation optimisation is to establish a usable measurement instrument for later experiments, rather than to optimise the subject model's routing directly.

1.4 Prospective representation probes

For a unit beginning at token index b_j, define a boundary representation $v_j^{(\ell)} = h_{b_j-1}^{(\ell)}$. Under causal processing, it depends on the problem and the preceding response, but not on the target unit. A binary linear probe for episode a has the form

$$\widehat{P}(a_j = a \mid v_j^{(\ell)}) = \sigma\left(w_{a,\ell}^{\intercal} v_j^{(\ell)} + c_{a,\ell}\right), \tag{1.7}$$

where σ is the logistic function. Above-chance prediction indicates that information useful to this classifier is accessible at that boundary. The interpretation depends on probe capacity, train/test separation, and possible shortcuts [11].

The `probeTest` experiment trains a separate one-versus-rest logistic regression for each label and hidden-state index. It teacher-forces released DeepSeek-R1 responses through a Qwen encoder. Consequently, its immediate subject is Qwen's representation of those supplied prefixes. It does not observe Qwen generating the annotated solutions. The current replication uses a sentence-level split, so units from a response can appear in both partitions; selecting the best layer on that test split also makes the selected maximum optimistic. A response-grouped evaluation and a separate layer-selection partition are needed to assess generalisation to new traces.

Temporal availability and causal influence must also be distinguished. A pre-unit representation is prospective, but predicting a future label does not establish that the probed direction causes that label. A probe can exploit problem type, position, or formatting. Independently trained, class-balanced binary scores also need not form calibrated multiclass probabilities, even when their largest value is used to propose a label for inspection.

Finally, hidden-state indices must be tied to actual locations in the architecture. Let $h_t^{(\ell)}$ denote a block-input residual state and $u_t^{(\ell)}$ the router input from Section 1.2. An intervening sequence-mixing sublayer and normalisation can make these vectors different. The current correlation extraction retains block-input hidden states alongside the router distribution from the corresponding block; it does not directly capture the exact normalised router-input vector. Geometric relationships between the two measurements must be interpreted with this distinction in mind.

1.5 Routing measurements and statistical analysis

1.5.1 Concentration, selection, and temporal dynamics

For a fixed layer and token, suppress the indices on p, and write $p_{(1)} \geq \cdots \geq p_{(E)}$ for the sorted router probabilities. For $E > 1$, Shannon entropy and its normalised concentration transform are

$$H(p) = -\sum_{e=1}^{E} p_e \log p_e, \qquad C(p) = 1 - \frac{H(p)}{\log E}. \tag{1.8}$$

$C = 0$ for a uniform distribution and approaches one as mass concentrates on one expert. Although the implementation calls C routing confidence, it is not a calibrated probability that the answer is correct. Routing probabilities describe expert allocation; vocabulary probabilities describe the next token. Their entropies concern different distributions.

For top-k routing with $k < E$, additional measures are

$$M_k(p) = \sum_{i=1}^{k} p_{(i)}, \tag{1.9}$$

$$B_k(p) = p_{(k)} - p_{(k+1)}, \tag{1.10}$$

$$G_k(p) = \frac{M_k(p)}{k} - \frac{1 - M_k(p)}{E - k}. \tag{1.11}$$

Selected mass M_k measures how much probability is assigned to the active set. The boundary margin B_k describes the separation between the last selected and first unselected expert. G_k compares mean selected and unselected probability. The top-1/top-2 margin alone does not describe the selection boundary when more than one expert is active.

Some of these quantities contain redundant information. For fixed E and k, G_k is an affine function of M_k; their associations cannot be treated as independent evidence. Similarly, C is an affine sign reversal of H. Presenting a transformed measure can aid interpretation, but does not create an additional empirical finding.

For a trace of $T > 1$ tokens, the current implementation summarises temporal routing with top-1 switching and top-k set overlap:

$$W^{(\ell)} = \frac{1}{T-1} \sum_{t=2}^{T} \mathbf{1}\left[\arg\max_{e} p_{t,e}^{(\ell)} \neq \arg\max_{e} p_{t-1,e}^{(\ell)}\right], \tag{1.12}$$

$$O^{(\ell)} = \frac{1}{T-1} \sum_{t=2}^{T} \frac{|S_{t-1}^{(\ell)} \cap S_{t}^{(\ell)}|}{k_\ell}. \tag{1.13}$$

The overlap denominator is the number of selected experts, not the union size used by Jaccard similarity. High overlap indicates persistent selection, but its relation to useful reasoning is an empirical question.

Identity features can instead measure the proportion of tokens selecting a particular (ℓ, e), or the frequency of a particular expert pair. An ordered top-1/top-2 pair describes co-selection at one token; it does not describe a transition between tokens or every combination in a top-k set. Layer identity must be preserved before such features can nominate individual experts for intervention.

Hidden-state summaries provide complementary measurements, including vector norm and consecutive-token cosine distance,

$$D^{(\ell)} = \frac{1}{T-1} \sum_{t=2}^{T} \left(1 - \frac{\langle h_t^{(\ell)}, h_{t-1}^{(\ell)} \rangle}{\|h_t^{(\ell)}\| \, \|h_{t-1}^{(\ell)}\|}\right). \tag{1.14}$$

The pipeline also constructs a geometry feature by correlating pairwise cosine similarities among sampled hidden states with the corresponding similarities among router distributions. That internal construction uses Pearson correlation. It is a scalar descriptive feature, distinct from the coefficient subsequently used to associate it with correctness; the many token pairs are not treated as independent experimental observations.

1.5.2 Binary correctness and repeated-attempt accuracy

Let $Y_{ia} \in \{0, 1\}$ indicate whether attempt a on problem i produces a correct final answer, and let X_{ia} be a routing or representation feature. Their attempt-level association is measured by point-biserial correlation,

$$r_{\mathrm{pb}} = \mathrm{Corr}(X, Y). \tag{1.15}$$

This is exactly Pearson correlation when correctness is encoded as zero or one. A positive coefficient means that the feature tends to be larger among correct attempts. Its magnitude can depend on class prevalence and feature distributions; it is neither a causal effect nor a measure of probability calibration. The same coefficient applies to a continuous feature paired with a binary reasoning-event indicator. Liu et al. provide direct language model precedent for this continuous-versus-binary setting [12].

With N attempts per problem, define

$$\widehat{A}_i^{(N)} = \frac{1}{N}\sum_{a=1}^{N} Y_{ia}, \qquad \overline{X}_i = \frac{1}{N}\sum_{a=1}^{N} X_{ia}. \tag{1.16}$$

The first quantity is the empirical per-problem avg@N. For $N = 32$, it takes values in $\{0, 1/32, \ldots, 1\}$. It differs from pass@N, which concerns at least one successful attempt among N samples. The primary repeated-attempt analysis uses one pair $(\overline{X}_i, \widehat{A}_i^{(N)})$ per problem and computes Spearman correlation,

$$\rho_s = \mathrm{Corr}\Big(\mathrm{rank}(\overline{X}_i), \mathrm{rank}(\widehat{A}_i^{(N)})\Big), \tag{1.17}$$

using average ranks for ties. The scientific question is whether larger feature values tend to accompany larger success fractions, without requiring a linear relation between their numerical scales. Boundedness alone does not make Pearson invalid; Spearman is selected because monotone association is the target. Small problem counts and many tied success fractions still limit precision. Section 2.6 discusses the supporting literature.

Complete groups are necessary to make avg@N comparable across problems. The analysis checks declared sample identifiers, outcome availability, and feature coverage, and audits exact duplicate generations where their fingerprints are available. A known gold answer with no parseable generated answer is counted as incorrect; a task without a scoring target remains unscored. Truncation and answer-parsing failures must be reported because they can alter both trace features and the measured success fraction.

1.5.3 Dependence, uncertainty, and confounding

Attempts on the same problem share difficulty and content. Counting them as independent observations can make uncertainty appear smaller than it is. For prespecified aggregate features, the pipeline therefore estimates 95% percentile bootstrap intervals by resampling whole problems and retaining their attempts together. For problem-level Spearman coefficients, it resamples the observed problem pairs. This latter procedure quantifies variation across observed problems without separately modelling uncertainty from the finite number of attempts within each problem.

An additional descriptive comparison uses only problems that yield both correct and incorrect attempts:

$$\Delta_i = \overline{X}_{i,Y=1} - \overline{X}_{i,Y=0}. \tag{1.18}$$

Averaging these contrasts removes fixed differences between source problems, but still permits differences in response length, token composition, or generation history within a problem. It also describes the subset of mixed-outcome problems. It is not an estimate of what would happen if routing were changed. The project's GPQA design varies answer-option positions between attempts, so it is excluded from this particular contrast.

Searching many layers, features, experts, and pairs creates a multiplicity problem. Benjamini–Hochberg adjustments provide a declared way to summarise multiple tests within a specified family [13]. They rely on valid input tests and appropriate dependence assumptions. In particular, applying the adjustment to naive attempt-level p-values does not repair their failure to account for repeated problems. Such values remain exploratory diagnostics. A confirmatory intervention should specify its outcomes, targets, and evaluation partitions independently of these scans.

1.6 From observations to routing intervention

A routing intervention modifies the computation in Equations 1.4–1.5 during inference. One possible family of policies adds a context-dependent bias to the router scores:

$$\widetilde{z}_t^{(\ell)} = z_t^{(\ell)} + \alpha\, v^{(\ell)}(q, x_{<t}), \tag{1.19}$$

after which expert selection and mixing weights are recomputed. This equation describes a design space for the thesis, rather than an already validated implementation. The strength α, affected layers, target experts, and activation times are parts of the policy.

Alternative policies reweight experts already selected, mask particular experts, or alter the selected set. These choices test different hypotheses. For example, rescaling all logits by a positive temperature changes their concentration but preserves their ordering and therefore preserves a deterministic top-k set. Increasing the contribution of an already selected expert also differs from forcing an otherwise unselected expert to participate. In all cases, renormalisation and the contribution of shared experts must be specified to interpret the intervention.

A controller intended to act before a reasoning unit can use only information available at that time. Offline labels may help construct or evaluate that controller, but the target unit and its future continuation cannot supply its decision at deployment. Prospective probes are relevant because they test whether useful information exists under this restriction. Their predictions still require validation on the generation distribution where the controller will operate.

Teacher-forced replay is useful for measuring activations on a fixed saved response, but an intervention during free generation changes subsequent tokens and therefore subsequent states. A replay-based change in a feature or next-token score is not sufficient evidence of improved problem solving. The intervention objective ultimately concerns a contrast such as

$$\Delta_\pi = \mathbb{E}_{q,\xi}\left[Y\big(q,\xi;\pi\big) - Y\big(q,\xi;\pi_0\big)\right], \tag{1.20}$$

where π_0 is the unmodified router, π is the intervention, and ξ represents sampling randomness under a specified evaluation protocol. Episode frequencies, answer correctness, and computational cost should be measured together: increasing the amount of checking language is not itself evidence of better reasoning.

This formulation motivates four connected research questions. Can the episode vocabulary be applied reliably to new traces? Can relevant behaviours be detected from information available before they occur? Which routing measurements remain associated with behaviour and correctness after accounting for major confounders? Finally, does acting on selected routing targets improve outcomes on held-out problems under a controlled generation budget? The three active experiments address the measurement and observational parts of this programme and establish the basis for answering the final, causal question.

Chapter 2

Related Work

Research relevant to routing-guided reasoning spans several levels of model behaviour. Some approaches change the prompts or training signals that produce reasoning traces. Others annotate those traces, decode information from internal representations, or directly modify activations and expert routing. This chapter compares those approaches according to the object they measure or manipulate, and positions the three active experiments within that literature.

2.1 Reasoning traces, evaluation, and functional annotation

Chain-of-thought prompting demonstrated that intermediate reasoning demonstrations can improve language-model performance on arithmetic, commonsense, and symbolic tasks [1]. This established generated reasoning text as a practical object of study, but it did not make final accuracy a complete measure of the quality of a reasoning process. Lightman et al. distinguish outcome supervision from feedback on intermediate step validity and release PRM800K for studying process supervision [14]. Their distinction is relevant to this thesis because a correct final answer and a well-formed intermediate process are related but separate evaluation targets.

Functional episode annotation addresses a further question: what a step is doing within the solution. Building on Schoenfeld's framework, Li et al.

annotate model-generated mathematical solutions at sentence and paragraph levels and analyse transitions between episode categories [9]. Their sentence-level labels supply the project's annotation vocabulary and released reference data. A semantic episode differs from a process-reward label: Verify describes an attempt to check something, whereas a validity label evaluates whether a step is correct. The same functional category can therefore occur in both successful and failed traces.

The faithfulness experiments of Turpin et al. further limit what can be inferred from generated explanations [2]. They show settings where input cues influence a prediction without being acknowledged in the accompanying rationale. For the present project, the implication is methodological: an episode sequence is an informative description of generated text, but a claim about the mechanism producing it needs independent internal evidence and intervention.

SPIRAL studies reasoning through a different route, using self-play on zero-sum games as a reinforcement-learning training signal [15]. Its evaluation includes mathematical and knowledge-intensive benchmarks also used by the correlation pipeline, and it analyses textual reasoning patterns. The project adopts this benchmark suite with its own benchmark-specific repeated-attempt protocol; it does not claim to reproduce every evaluation setting of the SPIRAL paper. Self-play changes the training process, whereas the thesis's intended control mechanism acts on routing during inference.

2.2 Prompt optimisation for episode annotation

An automatic episode annotator must handle closely related categories, class imbalance, and contextual ambiguity. GEPA addresses the broader problem of optimising language-model systems through reflective prompt evolution [10]. It uses execution and evaluation traces to generate natural-language feedback, proposes prompt revisions, and maintains candidates with complementary strengths through Pareto-based selection. The optimisation operates with fixed model weights.

The `gepaLLMAsJudge` experiment applies this approach to the seven-label annotation task. It supplies the problem and neighbouring text units, evaluates predictions against the released annotations, and preserves response boundaries in its data splits. A same-model judge can supply detailed feedback during the search, while agreement with the reference labels and per-class recall constraints govern final selection. These are properties of the project's experimental design, rather than claims about GEPA's original evaluation.

The distinction between optimisation feedback and final evaluation is central. A candidate selected after many comparisons on the same validation responses can exploit peculiarities of that partition. Furthermore, a judge that shares the classifier's model may share its errors. Response-grouped evaluation and an untouched test partition address a different question from whether the search found a better validation prompt. The resulting prompt is intended as a frozen annotation instrument for subsequent experiments; its offline context and reference-label provenance must remain visible when its labels are used to study routing.

2.3 Probing representations of reasoning

Probing asks whether a chosen decoder can extract a property from a model's representations. Hewitt and Manning demonstrate that a learned transformation of contextual word representations can recover aspects of syntactic tree geometry [16]. Hewitt and Liang subsequently show how control tasks help distinguish information in representations from a probe's capacity to learn the target task [11]. Together, these works motivate simple, inspectable probes and evaluation designs that make shortcut learning visible. Their linguistic tasks do not establish that reasoning episodes are encoded in the same way; that is an empirical question for this project.

Sun et al. study reasoning as trajectories through representation space [17]. Their analysis examines step-specific geometry and correctness signals in a Llama family, and they investigate adaptive activation changes toward reference trajectories. Their step categories describe positions in a sequence, rather than Schoenfeld semantic functions. This distinction makes their layerwise probing protocol relevant without making the two annotation tasks equivalent. They also report more limited transfer for correctness prediction than for step-related structure.

The `probeTest` experiment combines a boundary-probe design with the released semantic episode annotations. It extracts representations immediately before the target unit, which creates a prospective prediction task. Its current sentence-split replication is useful for comparison with the adopted protocol, but shared response identities and test-based layer selection limit generalisation claims. Moreover, applying the probes to supplied reference solutions tests representation transfer under teacher forcing. Using them to control the model's own generation requires additional validation.

This line of work supplies possible detection signals and candidate layers. It does not identify a router intervention by itself. A decoder may read information from a representation that is distributed across many experts, or from a component that the model does not use in the way the decoder does. The transition from decodability to controllability therefore remains a separate experimental step.

2.4 Expert specialisation and direct routing control

Early sparse MoE work establishes conditional computation as a way to scale capacity while limiting active computation [3]. Switch Transformer, Mixtral, and DeepSeekMoE explore different selection and expert-sharing designs [4, 5, 6]. These studies support the architectural motivation for treating routing as a control point. They also caution against assuming that its units already correspond to human-readable reasoning functions. In particular, Mixtral's analysis of syntactic routing and token locality makes lexical controls relevant when comparing experts across reasoning episodes.

RICE is a close precedent for reasoning-directed expert intervention [18]. Wang et al. associate experts with reasoning-related marker tokens using normalised pointwise mutual information, then reinforce identified experts during inference. Their reinforcement multiplies an expert's mixture weight when that expert is already in the selected set; it does not necessarily make an unselected expert participate. The paper's description of cognitive experts is a working hypothesis. Reported sensitivity to reinforcement strength and normalisation, together with limited sampled transfer in one AIME comparison, makes clear that observed specialisation does not automatically imply a robust intervention benefit.

SteerMoE investigates behaviour-associated expert activation using contrasting inputs, and promotes or suppresses experts by modifying routing scores before top-k selection [19]. Its experiments concern behaviours such as document faithfulness and safety. This provides direct evidence that expert selection can be used as an inference-time control mechanism for the studied settings. Its behavioural targets differ from mathematical episode structure, so the transfer to reasoning-guided routing remains a question. The difference from RICE is also operational: changing scores before selection can change membership of the active set, whereas reinforcing an already selected expert primarily changes its contribution.

Yoon et al. examine whether the trained router selects favourable expert combinations by comparing standard routes with sampled alternatives at equal expert compute [20]. Their counterfactual analysis uses successful continuations and also studies a final-router update. This is relevant evidence that routing quality can be investigated separately from expert capacity. However, a next-token score on a supplied continuation is a proxy for free-generation success, and layerwise route comparisons do not by themselves identify an episode-level intervention effect.

These studies establish that routing control is an existing research direction. The question developed here is more specific: whether a validated semantic episode vocabulary, prospective internal signals, and outcome-associated routing measurements can jointly support interventions that improve reasoning on held-out problems. Demonstrating that conjunction requires measurements and evaluations beyond finding experts associated with a few textual markers.

2.5 Activation steering

Routing edits should also be compared with methods that act directly on representations. Contrastive Activation Addition constructs steering directions from differences between activations on contrasting examples and adds those directions during generation [21]. This changes the residual representation rather than explicitly specifying which experts are selected. It is consequently a useful conceptual baseline for separating the value of a behavioural direction from the value of acting at the router.

Hong et al. study a residual-stream direction associated with the balance between reasoning and memorisation [22]. Their analysis relates scalar projections to graded reasoning scores and includes interventions on the corresponding features. Sun et al.'s trajectory work similarly moves beyond representation analysis to interventions that use reference reasoning trajectories [17]. These papers show why identifying a direction and testing its manipulation belong to distinct stages of a causal investigation.

For an MoE subject, a residual edit may indirectly alter routing as well as other downstream computations. Conversely, a router edit changes the mixture of expert outputs and therefore the residual trajectory that follows. The two intervention families are connected, but their direct targets differ. A routing-focused thesis must specify that target and evaluate its effects under comparable generation conditions. A change in episode labels alone would leave open whether the method improved problem solving or only changed how the response was expressed.

2.6 Statistical evidence for metric selection

The choice of correlation coefficient follows the outcome scale and the scientific question. Liu et al. use point-biserial correlation between continuous quantities, including context length and answer probability, and binary correctness in figurative-language interpretation [12]. This is direct precedent for the attempt-level continuous-feature/0–1-outcome setting. The additional justification is algebraic: point-biserial is Pearson correlation under binary coding, with an interpretable direction given by the group difference.

For graded targets, Hong et al. report Spearman association between residual-stream projections and model-assigned reasoning scores [22]. Hewitt and Manning use rank correlation in evaluating representation-derived syntactic geometry [16]. Xu et al. relate language-model-derived conceptual representations to human behavioural ratings using Spearman correlation and employ bootstrap confidence intervals [23]. These are precedents for rank-based analysis of internal quantities, rather than direct studies of avg@32 correctness. The project's choice of Spearman for per-problem mean correctness is therefore justified by its monotone association target, not by claiming that these papers uniquely prescribe that coefficient.

Zhang et al. report both Pearson and Spearman correlations when comparing uncertainty-based hallucination scores with graded human judgements [24]. Their use of entropy-related quantities is relevant to this project's concentration features, although token uncertainty and expert-routing entropy refer to different distributions. Reporting both coefficients can answer distinct linear and monotone questions; it does not turn two analyses of the same data into independent replications.

For the correlation experiment, the sample hierarchy matters as much as the coefficient. Thirty-two attempts from a single problem provide repeated observations of that problem, rather than 32 independent problems. The problem-level analysis and cluster bootstrap described in Section 1.5 address that hierarchy. Xu et al.'s use of 10,000 bootstrap resamples in one analysis provides a precedent for reporting resampling uncertainty, but does not determine an appropriate resample count or sampling unit for a different experiment [23].

Benjamini and Hochberg's multiple-testing framework motivates declaring families of feature, layer, and expert comparisons [13]. Multiplicity adjustment cannot remove confounding, repair non-independent tests, or turn a discovered association into a causal effect. The active analysis therefore serves to prioritise hypotheses whose intervention effects must subsequently be tested on separate data.

Table 2.1: Role of each active experiment in preparing routing intervention. These are research functions, rather than claims of completed causal validation.

Experiment	Evidence sought	Role in intervention research
`gepaLLMAsJudge`	Agreement between automatic episode labels and reference annotations on separated responses.	Defines a reproducible behavioural measurement instrument.
`probeTest`	Predictability of upcoming episodes from pre-unit hidden states.	Nominates signals and locations for detecting relevant contexts during generation.
`correlation_pipeline`	Associations of routing and hidden-state features with correctness and reasoning events.	Nominates routing properties and expert-use patterns for controlled tests.

2.7 Position of the three active experiments

Table 2.1 relates the active experiments to the evidence required by the thesis objective. Their contributions are complementary: a reliable annotation instrument makes behavioural comparisons possible; a prospective probe tests whether suitable signals are available before a decision; and a routing analysis narrows the set of mechanisms worth changing.

The current workflows also leave specific connections to establish. The correlation pipeline's default reasoning-event indicators are lexical backtracking, contradiction, and self-correction markers; the frozen seven-category annotator has not yet replaced them. Probe-selected locations remain exploratory under the current split and selection protocol. The initial expert-identity scan pools equal numeric expert indices across retained layers, so a target-specific intervention requires analysis that preserves the full (ℓ, e) identity. These distinctions prevent the existing components from being mistaken for a completed episode-aware controller.

Generation and measurement provenance impose another limit. The correlation workflow generates with a quantised GGUF model and replays the saved response through a corresponding Hugging Face checkpoint using a different 4-bit representation. The replay provides measurements for that implementation on the supplied tokens; shared checkpoint lineage does not make its activations numerically identical to those of the generator. Any future intervention evaluation must specify the actual model and routing implementation being modified.

The contribution pursued by the thesis is thus an experimentally grounded connection between semantic reasoning behaviour and controllable sparse computation. Existing work motivates both the possibility of steering and the need for careful interpretation. The three active experiments establish the vocabulary, measurements, and candidate signals needed to test that connection; the decisive evidence will come from whether a specified routing policy changes reasoning and improves final outcomes on held-out problems.

Chapter 3

Method

Chapter 4

Experimental Setup

Chapter 5

Results

Chapter 6

Conclusions

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